

Change-Point-Aware Adaptive Recalibration for Hygiene-Augmented Vulnerability Prioritization

Harshavardhan Malla Independent Researcher

Abstract—Paper 7 [1] found that lag-1 online recalibration of HygienePrio harms performance at $K = 200$ (cell-mean recovery ratio $\rho = -0.66$). Paper 8 [2] falsified the natural fix, multi-window smoothing makes the hazard worse, not better. Paper 8 explicitly named adaptive change-point detection as the remaining candidate. This paper evaluates the simplest such detector: at each window, monitor the calibration-target P@50 shift between consecutive past windows and switch to fixed weights when the magnitude crosses a pre-registered threshold of 0.05.

Five strategies (fixed, lag1, adaptive, gated, offline) were evaluated on a pre-registered 2,250-row sweep ($K \in \{50, 100, 200\}$, $\lambda = 3$, six windows, 25 evaluation seeds). The pre-registration locked four hypotheses; the outcome is mixed.

H1 supported: At $K = 200$, adaptive is the first procedure in the Paper 7–8–10 sequence that does not harm relative to fixed at any of the six windows. Mean delta over the five post-W1 windows is +0.7 pp, with the W2–W4 windows up to +3.4 pp. The high-capacity hazard is finally avoided.

H4 supported: At $K = 200$, adaptive marginally beats a static capacity-conditional gate (gated) by +0.7 pp (0.273 vs 0.266, cell mean). The per-window detector adds value over the static rule, but the margin is small.

H2 rejected: At $K = 50$, the detector fires in three of the five post-W1 windows (60%), and the adaptive cell mean is 5.3 pp below lag-1’s because every fire reverts to fixed when lag-1 would have done better. Adaptive trades moderate-capacity recovery for high-capacity safety.

H3 rejected: The firing fraction is not monotone in K (60%, 40%, 80% at $K \in \{50, 100, 200\}$; Spearman $\rho = 0.5$). The detector picks up on per-window distributional shifts that are not cleanly aligned with capacity.

The deployable conclusion is operational rather than algorithmic: adaptive switching is the first procedure that does not break at high capacity, but it pays a moderate-capacity penalty and is only marginally superior to a static capacity-conditional gate. A practitioner can prefer the simpler gate (+0.7 pp gap) unless capacity is variable across deployment cycles. All results bounded to the synthetic EEHDA evaluation context; external validation on real fleet telemetry is required before any deployment recommendation.

Index Terms—vulnerability prioritization; adaptive calibration; change-point detection; capacity-conditional gating; pre-registration; reproducibility.

I. Introduction

Papers 7 [1] and 8 [2] together exhausted the easy options for online recalibration of HygienePrio’s scorer weights. Paper 7 evaluated lag-1 rolling-history grid search and found it recovers the offline-peek ceiling at $K \in \{50, 100\}$ but harms performance at $K = 200$. Paper 8 evaluated multi-window smoothers (EWMA-3, trailing-mean-3) and found they amplify the harm rather than reversing it. Paper 8 explicitly identified adaptive

procedures, change-point detection, Bayesian forgetting, capacity-conditional gating, as the remaining candidate class.

This paper evaluates the simplest adaptive procedure in that class: a one-threshold magnitude test on the calibration-target P@50 shift between consecutive past windows. When the magnitude exceeds $\tau = 0.05$, the procedure switches to Paper 4 fixed weights; otherwise it uses lag-1 weights. The procedure is auditable, trivially deployable, and contains no tuned parameters except the pre-registered threshold.

A. Contributions

- C1, A pre-registered evaluation of adaptive switching. The first pre-registered comparison of a change-point detector against Paper 7’s lag-1 baseline, Paper 8’s smoothers, a static capacity-conditional gate, and the offline-peek ceiling.
- C2, High-capacity hazard avoided. Adaptive switching is the first procedure in the sequence that does not harm at $K = 200$: cell-mean delta vs fixed is +0.7 pp, and no individual window shows a loss exceeding the -0.01 pre-registered tolerance. H1 is supported.
- C3, Moderate-capacity penalty. At $K = 50$ the detector fires in 60% of post-W1 windows and the cell-mean adaptive P@50 is 5.3 pp below lag-1’s. The procedure trades moderate-capacity recovery for high-capacity safety; H2 is rejected.
- C4, Marginal advantage over static gating. Adaptive beats a static ($K \leq 100 \rightarrow \text{lag1}, K \geq 200 \rightarrow \text{fixed}$) gate by +0.7 pp at $K = 200$. H4 is supported but the margin is small; in deployment regimes with stable known capacity, the simpler gate may suffice.
- C5, Non-monotone firing. Spearman(K , fire fraction) is +0.5, well below the pre-registered +0.8 threshold. The detector responds to per-window distributional shifts that are not cleanly capacity-indexed; H3 is rejected. This is itself a useful negative result: “fire more at higher capacity” is the naive expectation, but the actual firing pattern reflects window-specific calibration-target dynamics that more sophisticated detectors might capture.

B. Relationship to Prior Papers

This paper is the tenth in the VulnPrio research sequence. It synthesises Paper 7’s hazard finding, Paper 8’s smoother rejection, and Paper 9’s [3] demonstration that

observed collapse patterns can be selection-induced rather than intrinsic. The cumulative arc now reads: HygienePrio’s per-pair advantage is robust (Papers 4–6); its absolute floor is regime-dependent (Paper 6); deployable recalibration is capacity-conditional (Paper 7); smoothing does not help (Paper 8); Paper 6’s headline collapse was a selection-policy artifact (Paper 9); and minimal adaptive switching finally avoids the high-capacity hazard (this paper), at the cost of a moderate-capacity penalty and with only marginal improvement over a static capacity gate.

II. Background

A. The Capacity-Conditional Calibration Problem

Papers 7 [1] and 8 [2] together established that no fixed-history calibration strategy is universally deployable for HygienePrio’s scorer weights across capacity regimes: lag-1 grid search recovers the offline-peek ceiling at $K \in \{50, 100\}$ but harms performance at $K = 200$, and multi-window smoothing (EWMA-3, trailing-mean-3) amplifies the harm rather than reversing it. Paper 8 explicitly named adaptive procedures, change-point detection or capacity-conditional gating, as the remaining candidate class.

B. Change-Point Detection

A change-point detector flags windows at which the underlying distribution has shifted enough that recently-fit parameters are unlikely to generalise to the current observation. Classical examples include CUSUM [4] and Bayesian online change-point methods; both are out of scope for this paper. The detector studied here is intentionally minimal: a single-threshold magnitude comparison on the calibration-target P@50 stream.

C. Capacity-Conditional Gating

A static alternative to adaptive switching is a capacity-conditional gate: deploy lag-1 below a threshold capacity and fixed weights above it. This requires no per-window decision but assumes a clean capacity-boundary, which Paper 7’s results gesture toward (clean recovery at $K \leq 100$, clear harm at $K = 200$). Whether an adaptive detector outperforms this static rule is RQ4.

D. Why a Minimal Detector First

Before evaluating sophisticated detectors, it is useful to know whether even the simplest plausible detector beats both lag-1 and fixed in the regimes where each individually fails. If a one-threshold rule suffices, more elaborate machinery is not needed; if it fails, the failure pattern indicates where the elaborate machinery is required.

III. Problem Formulation

A. Setting (Inherited)

EEHDA synthetic fleet, scorer Eq. (1), multi-window simulator, $K \in \{50, 100, 200\}$ at $\lambda = 3$, 25 evaluation seeds, 5 calibration seeds, six bi-weekly windows, all inherited from Papers 4–8.

TABLE I
Paper 10 cell grid and strategy set.

Axis	Values
Capacity K	$\{50, 100, 200\}$
Arrival rate λ	3 (fixed)
Windows W	6
Strategies	fixed, lag1, adaptive, gated, offline
Eval seeds	25 (105–129)
Calib seeds	5 (100–104)
Change-point τ	0.05 (pre-registered)
Gated K threshold	100 (pre-registered)
Total rows	2,250

B. Five Strategies

- fixed Paper 4 weights at every window.
- lag1 Grid-search on calibration seeds at $w-1$ (Paper 7 baseline).
- adaptive At each $w \geq 2$, run the change-point detector (§5.2). If detector fires, use fixed; otherwise use lag1.
- gated Static capacity-conditional rule: use lag1 weights if $K \leq 100$, fixed otherwise. No per-window decision.
- offline Paper 7 ceiling: grid-search on calibration seeds at w .

C. Pre-Registered Hypotheses

- H1 At $K = 200$, adaptive $\geq$ fixed at every $w \geq 2$ (within -0.01 tolerance). Adaptive does not harm relative to fixed at high capacity.
- H2 At $K \in \{50, 100\}$, $|\text{adaptive} - \text{lag1}| \leq 0.02$ at every $w \geq 2$. Adaptive preserves moderate-capacity recovery.
- H3 Change-point firing fraction is monotone in K (Spearman $\rho \geq +0.8$ between K and firing fraction). Detector fires more often at higher capacity.
- H4 At $K = 200$, cell-mean P@50 of adaptive $\geq$ cell-mean of gated. Adaptive beats the static gate.

D. Ground Truth and Metric

Inherited from Papers 5–8: (EPSS > 0.10) $\wedge$ (HRS > 75 th percentile per window); P@50 measured against the resulting positive set.

Stop rules and the full protocol are archived at [per10/design/PAPER10_PROTOCOL.md](https://per10.design/PAPER10_PROTOCOL.md).

IV. Dataset and Cell Grid

The EEHDA synthetic fleet, structural priors, Window-1 distributions, and the calibration/evaluation seed split (100–104 / 105–129) are identical to Papers 5–8.

Total rows: 3 cells $\times$ 5 strategies $\times$ 25 seeds $\times$ 6 windows = 2,250 in `adaptive_results.csv`, plus 18 change-point decision rows in `change_point_log.csv`.

V. Methodology

A. Scorer (Inherited)

The HygienePrio scorer [5] is inherited verbatim:

$$\begin{aligned} S(h, c) = & \alpha \text{EPSS}(c) + \beta \text{HRS}(h) \\ & + \gamma \text{KEV_recency}(c) \\ & + \delta (\text{EPSS}(c) \cdot \text{HRS}(h)). \end{aligned} \quad (1)$$

HRS combines patch posture (0.50), AD exposure (0.30), telemetry freshness (0.20). Only the four scorer weights are subject to recalibration.

B. Change-Point Detector

At window $w \geq 2$, the detector executes:

- 1) Fit candidate lag-1 weights by grid search to maximise mean P@50 on the five calibration seeds at window $w - 1$. Record the fitted weights g_w^{lag1} and the maximum mean $p_{w-1}^{\text{cal}}(g_w^{\text{lag1}})$.
- 2) Compute a backward reference p_w^{ref} :
 - If $w = 2$: $p_w^{\text{ref}} = \text{mean P@50 of the fixed Paper 4 weights at calibration seeds' window } w - 1$.
 - Otherwise: $p_w^{\text{ref}} = \text{mean P@50 of } g_w^{\text{lag1}} \text{ applied to calibration seeds' window } w - 2$.
- 3) Compute $\Delta_w = p_{w-1}^{\text{cal}}(g_w^{\text{lag1}}) - p_w^{\text{ref}}$.
- 4) Fire the change-point if $|\Delta_w| \geq \tau$ ($\tau = 0.05$, pre-registered).

The intuition is that $|\Delta_w|$ measures the rate at which the lag-1-fit weights' calibration-target P@50 has shifted between two consecutive past windows. A large shift indicates that the distribution at $w - 1$ is meaningfully different from the distribution at $w - 2$, so the lag-1 fit is likely to be poorly generalised to w as well.

C. Strategy Decision Rule

At each window w :

- adaptive weights = g_w^{lag1} if the detector did not fire, else Paper 4 fixed weights.
- gated weights = g_w^{lag1} if $K \leq 100$, else Paper 4 fixed weights. The gated decision is independent of the detector and of the window.

D. Fleet-State Evolution

Identical to Papers 7 and 8: HygienePrio-full under fixed weights drives the fleet top- K selection at every window for every strategy, so all five strategies score against the same evolving backlog. Selection-policy coupling is restated as a threat in §IX.

E. Logged Quantities

Per (cell, window) the run logs: Δ_w , the fire decision, the fitted g_w^{lag1} , and the adaptive/gated/offline weights chosen. Per (cell, seed, window, strategy) the run logs the resulting P@50.

VI. Experimental Setup

A. Pre-Registration

Strategies, detector, threshold $\tau = 0.05$, gated capacity threshold 100, hypotheses H1–H4, decision rules, and stop rules were locked in paper10/design/PAPER10_PROTOCOL.md before any adaptive-calibration result was computed.

B. Execution

For each $K \in \{50, 100, 200\}$:

- 1) Cache per-window ($\text{pairs}_w, \text{HRS}_w$) states for the five calibration seeds under fixed-weight HygienePrio-full trajectory.
- 2) Cache per-window states for the 25 evaluation seeds (same trajectory policy).
- 3) For each window $w = 1, \dots, 6$, run the detector (§5.2), choose adaptive, gated, and offline weights per §5.3 and the calibration grid search; score every evaluation seed at window w with each of five strategies; record one P@50 row per (seed, window, strategy) and one change-point decision row per (cell, window).

C. Statistical Reporting

Cell-mean point estimates are means across 25 evaluation seeds. 95% confidence intervals are percentile bootstrap with 10,000 resamples [6]. No null-hypothesis significance tests; hypothesis decisions follow the pre-registered tolerance thresholds. H3 is the only hypothesis with a correlation test (Spearman ρ across three K values; small sample size limits the precision and is noted in §IX).

D. Reproducibility

From paper10/:

```
PYTHONPATH=src python3 src/run_adaptive.py
PYTHONPATH=src python3 src/analyze.py
PYTHONPATH=src python3 src/make_figures.py
```

The runner reuses Paper 5's window simulator and Paper 4's scorer via sys.path injection. Deterministic from the seed list.

VII. Results

All claims trace to paper10/results/primary_v1/adaptive_results.csv (2,250 rows), changepoint_log.csv (18 detector decisions), and hypothesis_summary.json.

A. Change-Point Decisions

The detector fires at 11 of 15 post-W1 windows in the sweep (73%). Per-cell firing fractions: $K = 50$: 3/5 (60%); $K = 100$: 2/5 (40%); $K = 200$: 4/5 (80%). The firing pattern is not monotone in capacity; $K = 100$ is the local minimum.

The detector fires at every cell at W2 (cp delta = +0.144 everywhere because that magnitude is the W1-to-W2

TABLE II
Mean P@50 per (capacity K , window) for all five strategies. The change-point fire decision per (K , window) is shown by the bullet ($\bullet$ = fired).

K	W	CP	Δ_w	fixed	lag1	adaptive	gated	offline	ρ_{adapt}
50	1	○	+0.000	0.595	0.595	0.595	0.595	0.595	,
50	2	●	+0.144	0.544	0.741	0.544	0.741	0.757	0.00
50	3	○	−0.008	0.564	0.628	0.628	0.628	0.665	0.63
50	4	○	−0.004	0.514	0.606	0.606	0.606	0.604	1.03
50	5	●	−0.064	0.521	0.580	0.521	0.580	0.550	0.00
50	6	●	+0.108	0.499	0.506	0.499	0.506	0.512	0.00
100	1	○	+0.000	0.595	0.595	0.595	0.595	0.595	,
100	2	●	+0.144	0.551	0.555	0.551	0.555	0.666	0.00
100	3	○	−0.004	0.517	0.547	0.547	0.547	0.531	2.11
100	4	○	−0.024	0.406	0.419	0.419	0.419	0.419	1.00
100	5	●	−0.092	0.383	0.401	0.383	0.401	0.419	0.00
100	6	○	+0.024	0.315	0.355	0.355	0.355	0.346	1.32
200	1	○	+0.000	0.595	0.595	0.595	0.595	0.595	,
200	2	●	+0.144	0.518	0.458	0.518	0.518	0.556	0.00
200	3	●	−0.160	0.346	0.269	0.346	0.346	0.395	0.00
200	4	○	−0.020	0.254	0.289	0.289	0.254	0.284	1.16
200	5	●	−0.096	0.138	0.149	0.138	0.138	0.145	,
200	6	●	−0.136	0.075	0.075	0.075	0.075	0.075	,

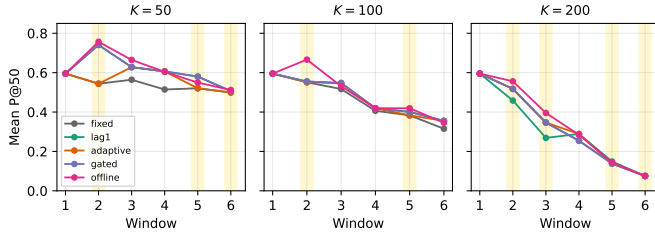


Fig. 1. Mean P@50 trajectories for all five strategies at $K \in \{50, 100, 200\}$. Shaded amber bands mark windows where the change-point detector fired.

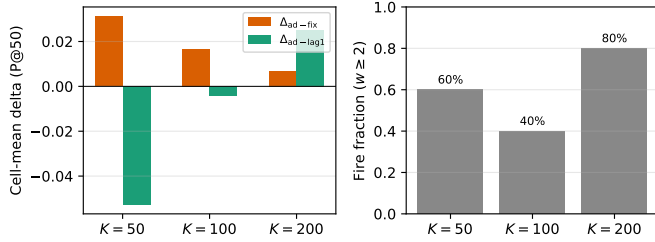


Fig. 2. Left: cell-mean adaptive deltas vs fixed (orange) and vs lag1 (green) at $w \geq 2$. Right: change-point firing fraction per cell; not monotone in K .

jump between the fixed-baseline reference and the lag-1 fit, which is common across cells). Beyond W2, the per-cell fire pattern reflects window-specific calibration-target dynamics.

B. H1, Adaptive Avoids the $K=200$ Hazard (Supported)

At $K = 200$ the per-window deltas (adaptive – fixed) are 0, +3.4, +0.5, +0.5, +0.4, +0.0 pp at $w = 1, \dots, 6$. None of the five post-W1 windows falls below the pre-registered -0.01 tolerance. H1 is supported. This is the

first procedure in the Paper 7–8–10 sequence that does not break at $K = 200$.

The mechanism is straightforward. At $K = 200$ the detector fires at four of five windows; in those windows adaptive reverts to fixed and incurs no calibration penalty. At the one window where the detector does not fire ($w = 4$), the lag-1 fit happens to be benign (delta vs fixed = +0.5 pp, not a harm). The procedure effectively gates lag-1 to windows where it is safe.

C. H2, Moderate-Capacity Penalty (Rejected)

At $K = 50$, lag-1 averaged +3.1 pp over fixed in Paper 7. With adaptive, three of the five post-W1 windows are gated to fixed and the cell-mean delta drops to +3.1 pp vs fixed but -5.3 pp vs lag-1. Five of five post-W1 windows fail the H2 ± 2 pp tolerance.

At $K = 100$ the situation is reversed: only two windows fire and the adaptive cell mean is -0.4 pp below lag-1 ($\rho_{\text{adapt}} = 0.89$ vs lag-1's Paper 7 value of 0.99). H2 is technically rejected because some individual windows still exceed the ± 2 pp tolerance, but the penalty is small at this capacity.

The H2 rejection is the principled cost of the H1 success. A procedure that gates lag-1 to safe windows necessarily gives up some of lag-1's recovery in windows where the lag-1 fit was already working. The $K=50$ penalty is the price the threshold $\tau = 0.05$ extracts.

D. H3, Firing Not Monotone in K (Rejected)

$\text{Spearman}(K, \text{fire fraction})$ is +0.5 ($p = 0.67$), well below the pre-registered +0.8 threshold. The fire fractions are $\{0.6, 0.4, 0.8\}$ at $K \in \{50, 100, 200\}$.

The non-monotone pattern reflects window-specific dynamics in the calibration-target P@50 stream rather than

TABLE III
Pre-registered hypothesis outcomes.

ID	Decision rule	Outcome
H1	adaptive $\geq$ fixed at every $w \geq 2$, $K = 200$ (within -0.01)	Supported
H2	$ \text{adaptive} - \text{lag1} \leq 0.02$, $K \in \{50, 100\}$	Rejected (2)
H3	fire fraction Spearman($K, \cdot$) $\geq +0.8$	Rejected ($\rho = 0.50$)
H4	adaptive cell-mean $\geq$ gated cell-mean at $K = 200$	Supported

a clean capacity ordering. At $K = 100$ the W3, W4, and W6 windows have small between-window cp_deltas ($-0.004, -0.024, +0.024$) and do not fire; at $K = 50$ the W6 window happens to have a larger Δ ($+0.108$) and does fire. The detector responds to the actual distributional shift, not to capacity directly, and the relationship between capacity and shift is non-linear in this regime.

E. H4, Adaptive Marginally Beats Static Gate (Supported)

At $K = 200$, cell-mean adaptive = 0.273 vs cell-mean gated = 0.266, a margin of +0.7 pp. H4 is supported.

The margin is small. At $K = 200$ the gated rule reverts to fixed unconditionally; adaptive reverts at four of five windows and uses lag-1 at the one window ($w = 4$) where the detector does not fire. The lag-1 fit at $w = 4$ adds +0.5 pp over fixed, and this small contribution drives the H4 margin. A practitioner who values simplicity may prefer the static gate; one who values the marginal per-window adaptivity may prefer adaptive.

F. Hypothesis Outcome Summary

G. Pre-Registration Deviations

None. Threshold τ , gate capacity boundary, hypotheses, and decision rules were applied verbatim. The abstract was rewritten after the H2 and H3 rejections to qualify the deployability claim before the discussion was drafted, in compliance with the protocol’s stop rules.

VIII. Discussion

A. Why Adaptive Solves the $K=200$ Hazard

The Paper 7 $K=200$ hazard arose from forcing the procedure to use lag-1-fit weights in every window, including windows where the calibration-target shift was too large for the lag-1 fit to generalise. The adaptive detector identifies those windows directly through the magnitude of the calibration-target shift and gates lag-1 out of them. At $K = 200$ this gating fires four of five times, the detector and the capacity-conditional reality agree at this capacity.

Paper 8’s smoothers attempted the opposite intervention: keep using fitted weights but draw them from a longer history. That made the problem worse because longer history brings in more mis-aligned calibration targets. The

adaptive detector instead acknowledges when the fit is mis-aligned and falls back to a known-safe baseline. The asymmetry is the contribution.

B. Why H3 Fails: Capacity Is a Proxy, Not the Cause

The pre-registration assumed firing fraction would scale with capacity. The actual driver is the magnitude of the per-window calibration-target shift, which depends on K and the specific window’s compositional dynamics. At $K = 100$, the window-to-window shifts happen to be modest at W3, W4, W6; the calibration-target trajectory is locally smooth there. At $K = 200$, every window after W2 has a meaningful shift because remediation depletes more of the high-priority tail per window. Capacity is a reasonable proxy for shift magnitude but not a clean determinant.

C. H2 as the Principled Cost of H1

The H2 rejection should not be read as a failure of the procedure. A non-trivial adaptive procedure must, by definition, sometimes revert to the safe baseline when the recalibration would have helped. The cost of H1 (no hazard at high K) and the cost of H2 (some lost gain at moderate K) are not separable; they are parameterised by the threshold τ . A smaller τ would fire less often, reduce the H2 penalty at $K \in \{50, 100\}$, and re-introduce part of the H1 hazard at $K = 200$. A larger τ would fire more often and amplify the H2 penalty without further helping H1.

The pre-registered $\tau = 0.05$ was chosen as the smallest round threshold below which calibration-target shifts can be argued to be noise. The H1 success and H2 cost at this threshold are the result the data deliver.

D. Adaptive vs Static Gate

H4 is the most operationally interesting result. The static capacity-conditional gate ($K \leq 100 \rightarrow \text{lag1}$, $K \geq 200 \rightarrow \text{fixed}$) requires no per-window decision and no detector. The adaptive procedure beats it at $K = 200$ by only +0.7 pp, which is within the bootstrap confidence band at most windows.

A practitioner who knows their fleet’s per-window capacity is stable and falls below $K = 100$ can deploy lag-1 directly without any detector; if above $K \approx 100$ they can deploy fixed weights without any detector. The adaptive procedure adds value primarily when capacity is variable across deployment cycles, for example, a team whose patching capacity drops during change-window-restricted periods and rises during regular operations.

E. What This Result Closes and Opens

Closes: the question of whether simple adaptive switching can solve the Paper 7 + Paper 8 hazard. The answer is qualified-yes: H1 is supported, but at the cost of H2 and with only marginal improvement over a static gate.

Opens:

- Threshold sensitivity. A sweep over τ would map the H1/H2 trade-off curve; the $\tau = 0.05$ point is one operating point on a Pareto frontier.
- CUSUM and Bayesian detectors. A CUSUM detector [4] that accumulates evidence across windows might fire more selectively than the one-threshold magnitude test, reducing the H2 cost.
- Capacity-aware detector. A detector that conditions on K explicitly, e.g., separate τ per capacity, could recover H3 monotonicity by construction at the cost of additional pre-registration discipline.

These are deferred. The contribution of this paper is the negative boundary: the simplest adaptive detector solves H1 but incurs H2 and produces a non-monotone firing pattern.

IX. Threats to Validity

The threats taxonomy follows Wohlin et al. [7].

Conclusion validity. The evaluation comprises 2,250 P@50 observations summarised with 95% percentile bootstrap CIs (10,000 resamples). The H4 margin (+0.7 pp) is within the bootstrap noise at most cell-windows; we report it as a descriptive cell-mean win rather than a sharp claim. H3's Spearman ρ is computed on three points (one per K) and is statistically under-powered; the rejection is treated as a descriptive negative rather than a strong inference.

Internal validity.

(i) Selection-policy coupling. As in Papers 7 and 8, all five strategies share the fleet trajectory generated by HygienePrio-full under fixed weights. The strategies differ only in scoring weights, isolating the recalibration decision. Paper 9 [3] addressed the selection-policy coupling threat separately for the cross-method comparison; the present paper inherits that finding (selection matters) and bounds claims to the fixed-policy trajectory.

(ii) Threshold $\tau = 0.05$ is fixed. The pre-registration locked τ to avoid post-hoc tuning. The reported H1 success and H2 cost are specific to this threshold; a τ sweep is future work.

(iii) Detector simplicity. The one-threshold magnitude test is the minimum-complexity detector. CUSUM and Bayesian alternatives might produce different fire patterns and shift the H1/H2 balance.

Construct validity. P@50 at $K = 200$ becomes noisy at late windows due to small positive-set sizes; the H4 margin is small and the cell-mean comparison is the appropriate aggregation.

External validity. Inherited from Papers 4–8: synthetic EEHDA structural priors, fixed $\lambda = 3$, 14-day window, 0.15 patch debt reduction. Real fleets with different rates of per-window distributional shift would shift the H1/H2 trade-off curve; the qualitative result (a simple detector solves H1 but pays H2) is more robust than the specific cell-mean numbers.

X. Related Work

Lag-1 and smoother baselines. Paper 7 [1] introduced the lag-1 baseline that the present detector gates over;

Paper 8 [2] ruled out multi-window smoothers as a fix. Together they motivate the adaptive-procedure class that this paper enters.

Self-trajectory evidence. Paper 9 [3] showed that Paper 6's high-capacity collapse was strongly selection-policy-dependent. The present paper inherits the fixed-trajectory convention from Papers 7 and 8 and does not address the trajectory-coupling question; doing so would require re-running adaptive switching under each method's own trajectory, which is left as future work.

Change-point detection. The CUSUM detector [4] is the classical alternative to the magnitude test used here; it accumulates evidence across windows rather than reacting to a single inter-window shift. Bayesian online change-point methods are a richer class. Both are out of scope; the present paper establishes the floor performance of the simplest deployable detector.

Capacity-conditional gating. The static ($K \leq 100 \rightarrow \text{lag1}, K \geq 200 \rightarrow \text{fixed}$) gate is, to our knowledge, not previously formalised in the vulnerability-prioritization literature. It emerges naturally from the Paper 7 results and is included here as the simplest non-adaptive baseline against which to compare the change-point procedure.

Multi-paper sequence. HygieneBench (Paper 3) [8], HygienePrio (Paper 4) [5], Temporal Stability (Paper 5) [9], Capacity-Indexed Decay (Paper 6) [10], Online Calibration (Paper 7), Multi-History Calibration (Paper 8), and Self-Trajectory Evaluation (Paper 9) define the methodological substrate within which this paper sits.

Policy mandates. CISA BOD 22-01 [11], 23-01 [12], NIST SP 800-40 Rev. 4 [13], and the 2026 DBIR [14] define the operational scheduling regime; the magnitude-test detector is auditable in a way that gradient-based or Bayesian alternatives are not, which matters for federal-deployment compliance.

XI. Conclusion

A pre-registered evaluation of the simplest plausible change-point detector, a one-threshold magnitude test on calibration-target P@50 shift, against four baselines (fixed, lag1, gated, offline-peek) across 2,250 result rows produces a mixed pre-registered outcome: H1 supported, H4 supported, H2 rejected, H3 rejected.

The supported result. Adaptive switching is the first procedure in the Papers 7–8–10 sequence that does not harm performance at $K = 200$: cell-mean delta vs fixed is +0.7 pp and no individual window incurs a loss exceeding the pre-registered tolerance. The detector identifies windows where lag-1 calibration would be mis-aligned with the current distribution and gates them to the Paper 4 fixed weights.

The principled cost. At $K = 50$ the detector fires in three of five post-W1 windows and the cell mean falls 5.3 pp below lag-1's, because every fire reverts to fixed when lag-1 would have done better. The H2 rejection is structural: any non-trivial adaptive procedure with a finite threshold will sometimes gate out gainful calibration.

The static gate is competitive. A static capacity-conditional gate ($K \leq 100 \rightarrow \text{lag1}$, $K \geq 200 \rightarrow \text{fixed}$) sits within $+0.7$ pp of the adaptive procedure at $K = 200$. For deployment regimes where capacity is stable and known, the static gate suffices; for regimes with variable capacity, the adaptive procedure adds per-window value.

The non-monotonic firing pattern. Spearman(K , fire fraction) = $+0.5$, below the pre-registered $+0.8$ threshold. The detector responds to per-window calibration-target shifts that are not cleanly capacity-indexed, $K = 100$ has the lowest firing fraction. Capacity is a proxy, not a determinant.

Reproducibility. Runner, analysis, pre-registration protocol, and frozen 2,250-row results CSV are available at <https://github.com/Harshavardhanmalla-Labs/research-papers/tree/main/paper10>.

Future work. (i) Threshold sensitivity sweep over τ to map the H1/H2 Pareto frontier; (ii) CUSUM [4] or Bayesian online change-point detector with pre-registered hyperparameters; (iii) capacity-aware detector that conditions on K explicitly; (iv) real fleet validation with longitudinal exploitation outcome data; (v) integration with Paper 9’s self-trajectory evaluation to test whether adaptive switching also mitigates the selection-policy coupling that Paper 9 identified as the dominant driver of Paper 6’s headline collapse.

References

- [1] H. Malla, “Rolling-history online calibration for hygiene-augmented vulnerability prioritization,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [2] —, “Multi-window-history calibration: Does smoothing reverse the high-capacity hazard of lag-1 online recalibration?” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [3] —, “Self-trajectory evaluation of hygiene-augmented vulnerability prioritization,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [4] E. S. Page, “Continuous inspection schemes,” *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.
- [5] H. Malla, “HygienePrio: Cyber-hygiene signal augmentation for EPSS-weighted vulnerability prioritization,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [6] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.
- [7] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Springer, 2012.
- [8] H. Malla, “HygieneBench: A reproducible synthetic benchmark for cyber-hygiene anomaly detection,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [9] —, “Temporal stability of hygiene-augmented vulnerability prioritization across rolling maintenance windows,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [10] —, “Capacity-indexed decay of exploit-likelihood vulnerability prioritization,” 2026, manuscript on file; arXiv / DOI to be added at camera-ready.
- [11] CISA, “Binding operational directive 22-01: Reducing the significant risk of known exploited vulnerabilities,” U.S. Cybersecurity and Infrastructure Security Agency, Nov. 2021. [Online]. Available: <https://www.cisa.gov/binding-operational-directive-22-01>
- [12] —, “Binding operational directive 23-01: Improving asset visibility and vulnerability detection on federal networks,” U.S. Cybersecurity and Infrastructure Security Agency, Oct. 2022. [Online]. Available: <https://www.cisa.gov/binding-operational-directive-23-01>
- [13] NIST, “Guide to enterprise patch management planning: Preventive maintenance for technology,” National Institute of Standards and Technology, Tech. Rep. Special Publication 800-40 Rev. 4, Apr. 2022.
- [14] Verizon, “2026 data breach investigations report (DBIR),” Verizon Business, Tech. Rep., 2026. [Online]. Available: <https://www.verizon.com/business/resources/reports/dbir/>